National University of Singapore
School of Computing
CS3244: Machine Learning
Solution to Tutorial 3 (2 weeks)

SVM and Kernelization

Readings: *Learning From Data*, Chapter 8 (Dynamic e-Chapters) – SVM and Kernelization

Part 3A (In-class)

1. **SVM**: Consider a binary classification problem. Let us denote the input set as $\{\mathbf{x}_n, y_n\}$ where $\mathbf{x}_n$ is the data point and $y_n \in \{-1, 1\}$ is the corresponding class label. The constraint of the SVM classifier can be given by:

$$y_n(W^t\Phi(\mathbf{x}_n) + b) \geq 1$$

Show that, if the 1 on the right-hand side of the constraint is replaced by some arbitrary constant $\gamma > 0$, the solution for the maximum margin hyperplane remains unchanged.

Then we have the following equation:

$$y_n(W^t\Phi(\mathbf{x}_n) + b) \geq \gamma$$

SVM construction: Remember the construction of the SVM: We want to find a boundary between the two classes such that the minimum distance of a data point $\mathbf{x}_n$ *to the boundary should be maximized.*

Construction: Distance of $\mathbf{x}_n$ *from the boundary is given by* $\frac{1}{||w||}[y_n(W^t\Phi(\mathbf{x}_n) + b)]$

So the optimization function we want to solve:

$$arg\max_{w,b}\left\{\frac{1}{||W||}\min_n[y_n(W^t\Phi(\mathbf{x}_n) + b)]\right\}$$

A direct optimization of this function is difficult, and as we have done in lecture we will convert it into an equivalent problem that is much easier to solve. To do this, we note that if we make the rescaling $W \to \gamma W$ *and* $b \to \gamma b$*, the distance will remain unchanged. This gives us the freedom to choose the* ***minimum distance constraint****:*

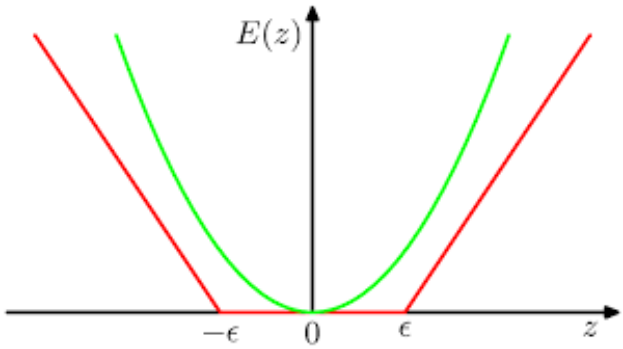
$$y_n(W^t\Phi(\mathbf{x}_n) + b) = \gamma$$

This will simply rescale $W \to \gamma W$ *and* $b \to \gamma b$*.*

2. **Support Vector Regression**: The support vector machinery generates a linear model, with further constraints. We can meld linear regression and SVM together to form a hybrid of support vector based regression algorithms. Let us replace the loss function in the SVM with ϵ-insensitive error function as follows:

$$E_\epsilon(g(\mathbf{x}) - y) = \begin{cases} 0 & \text{if } |g(\mathbf{x}) - y| < \epsilon, \\ |g(\mathbf{x}) - y| - \epsilon & \text{otherwise.} \end{cases}$$

where $g(\mathbf{x})$ is the predicted value for input $\mathbf{x}$ and y is the corresponding target. This loss function is shown in Figure 1. Can you extend the idea of SVM to formulate the optimization function for this regression problem? *Hint: Apply slack variables to obtain the optimization formulation.*



(**Figure 1.** Illustration of squared loss (green) and ϵ-insensitive loss (red), taken from `http://www.robots.ox.ac.uk/~az/lectures/ml/lect4.pdf`. The y-axis denotes the loss value and x-axis the quantity $|g(\mathbf{x} - y|$.)

The (regularized) error function can be given by:

$$C \sum_n E_\epsilon(g(\mathbf{x}) - y) + \frac{1}{2}||W||^2$$

Here C is the user-given (regularizer) parameter. Note that, this optimization problem is difficult to solve and we need to replace the error term with constraints as follows:

According to our error function, each of the target values y_n should fall within $g(\mathbf{x}_n) - \epsilon$ and $g(\mathbf{x}_n) + \epsilon$. Otherwise, we will incur an error.

Now, for soft margin case, we introduce two slack variables $\xi_n \geq 0$ and $\hat{\xi}_n \geq 0$ for each data point $\mathbf{x}_n$. We can then rewrite the error function as follows:

$$y_n \leq g(\mathbf{x}_n) + \epsilon + \zeta_n$$

$$y_n \geq g(\mathbf{x}_n) - \epsilon - \hat{\zeta}_n$$

Then the error we want to minimize for each data point $\mathbf{x}_n$ is $\xi_n + \hat{\xi}_n$. So the total error is $\sum_n [\xi_n + \hat{\xi}_n]$.

Hence the optimization we want to solve:

$$\min \frac{1}{2}||w||^2 + \sum_n [\xi_n + \hat{\xi}_n]$$

Such that:

$$y_n \leq g(\mathbf{x}_n) + \epsilon + \xi_n$$
$$y_n \geq g(\mathbf{x}_n) - \epsilon - \hat{\xi}_n$$
$$\xi_n, \hat{\xi}_n \geq 0 \quad \forall n$$

Kernelization: Kernelization is an important byproduct of the development of support vector machines. The idea that a kernel function that represents a dot product in some space can be applied to the Gram matrix calculations in the quadratic programming problem in support vector machines, but can be leveraged in other places in learning algorithms. It is important to understand the properties of kernels.

3. By considering the determinant of a 2×2 Gram matrix, show that a positive semidefinite kernel function $K(\mathbf{x}, \mathbf{x}')$ satisfies the inequality

$$K(x_1, x_2)^2 \leq K(x_1, x_1)K(x_2, x_2)$$

Consider the two-dimensional scenario i.e $\boldsymbol{x} = \{x_1, x_2\}$. *Then the* 2×2 *gram matrix for* $K(\mathbf{x}, \mathbf{x})$ *is given by:*

$$\begin{bmatrix} K(x_1, x_1) & K(x_1, x_2) \\ K(x_2, x_1) & K(x_2, x_2) \end{bmatrix}$$

The determinant of this matrix can be given as :

$$K(x_1, x_1)K(x_2, x_2) - K(x_1, x_2)K(x_2, x_1)$$

Now, by definition a kernel is a symmetric function, i.e. $K(x_1, x_2) = K(x_2, x_1)$. *Additionally, we know a kernel has the positive semidefinite property: i.e., the determinant is non-negative.*

$$K(x_1, x_1)K(x_2, x_2) - K(x_1, x_2)K(x_2, x_1) \geq 0$$

$$K(x_1, x_1)K(x_2, x_2) \geq K(x_1, x_2)^2$$

4. A kernel is valid if it corresponds to a scalar product in some (perhaps infinite dimensional) feature space. Remember a necessary and sufficient condition for a function $K(-,-)$ to be a valid kernel is that the Gram matrix, whose elements are given by $K(\mathbf{x}_n, \mathbf{x}'_m)$, should be positive semi-definite for all possible choices of the set X. Test the the kernel:

$$K(\mathbf{x}, \mathbf{x}') = \langle \mathbf{x}, \mathbf{x}' \rangle^3 + (\langle \mathbf{x}, \mathbf{x}' \rangle + 1)^2$$

where $\langle -, - \rangle$ indicates an inner product. *Hint: Check how two construct a valid kernel by combining one or more valid kernels.*

First notice that $\langle \mathbf{x}, \mathbf{x}' \rangle$ *is a valid kernel as it is a scalar product in the Euclidean space. Next we will use some properties of constructing valid kernel:*

<u>Property 1</u>: given two valid kernels $K_1(-,-)$ *and* $K_2(-,-)$*:*

$$K(\mathbf{x}, \mathbf{x}') = K_1(\mathbf{x}, \mathbf{x}') + K_2(\mathbf{x}, \mathbf{x}')$$

Then we just need to show: $\langle \mathbf{x}, \mathbf{x}' \rangle^3$ *and* $(\langle \mathbf{x}, \mathbf{x}' \rangle + 1)^2$ *both are separately valid kernels. Proof for* $\langle \mathbf{x}, \mathbf{x}' \rangle^3$.

Property 2: given two valid kernels $K_1(-,-)$ *and* $K_2(-,-)$:

$$K(\mathbf{x}, \mathbf{x}') = K_1(\mathbf{x}, \mathbf{x}') * K_2(\mathbf{x}, \mathbf{x}')$$

Now, we know that $\langle \mathbf{x}, \mathbf{x}' \rangle$ *is a valid kernel. Hence* $\langle \mathbf{x}, \mathbf{x}' \rangle^2$ *and so is* $\langle \mathbf{x}, \mathbf{x}' \rangle^3$.

Proof of $(\langle \mathbf{x}, \mathbf{x}' \rangle + 1)^2$

Property 3: given two valid kernels $K_1(-,-)$:

$$K(\mathbf{x}, \mathbf{x}') = q[K_1(\mathbf{x}, \mathbf{x}')]$$

where $q[.]$ *is a polynomial function with a non-negative coefficients.*

Again we know that $\langle \mathbf{x}, \mathbf{x}' \rangle$ *is a valid kernel. Hence* $\langle \mathbf{x}, \mathbf{x}' \rangle^0$ *is also a valid kernel. Thus,* $(\langle \mathbf{x}, \mathbf{x}' \rangle + 1)$ *is also valid and so is* $(\langle \mathbf{x}, \mathbf{x}' \rangle + 1)^2$.

Note that we can simply apply Property 3 to prove these above statement.

Part 3B (Webcast)

1. **SVM:** Recall that the equation of an ellipse in the 2-dimensional plane is:

$$c(x_1 - a)^2 + d(x_2 - b)^2 - 1 = 0$$

Show that an SVM using the polynomial kernel of degree $Q = 2$, $K(\mathbf{x}, \mathbf{x}') = (1 + \langle \mathbf{x}, \mathbf{x}' \rangle)^2$, is equivalent to a linear SVM in the feature space $\{1, x_1, x_2, x_1^2, x_2^2, x_1 x_2\}$, for 2-dimensional (i.e., $d = 2$) vectors and hence that SVMs with this kernel can separate any elliptic region from the rest of the plane.

Remember the condition of function being a valid kernel: given a function of two vector, if we can express that function as a dot product in a higher dimensional space.

According to the question, $\mathbf{x}$ *and* $\mathbf{x}'$ *are two dimensional vectors. To prove the first part of the question, we need to expand the kernel and write it in terms of dot product in the specified feature space.*

$$\begin{aligned} K(\mathbf{x}, \mathbf{x}') &= (1 + \langle \mathbf{x}, \mathbf{x}' \rangle)^2 \\ &= (1 + x_1 x_1' + x_2 x_2')^2 \\ &= 1 + 2x_1 x_1' + 2x_2 x_2' + 2x_1 x_1' x_2 x_2' + (x_1 x_1')^2 + (x_2 x_2')^2 \\ &= (1, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1 x_2, x_1^2, x_2^2).(1, \sqrt{2}x_1', \sqrt{2}x_2', \sqrt{2}x_1' x_2', x'^2_1, x'^2_2) \end{aligned}$$

Hence, we can express the kernel in the specified feature space (ignore the constant terms i.e $\sqrt{2}$*).*

Now, for the proof of the second part, we can expand the ellipse equation and show that any ellipse can be expressed in this feature space.

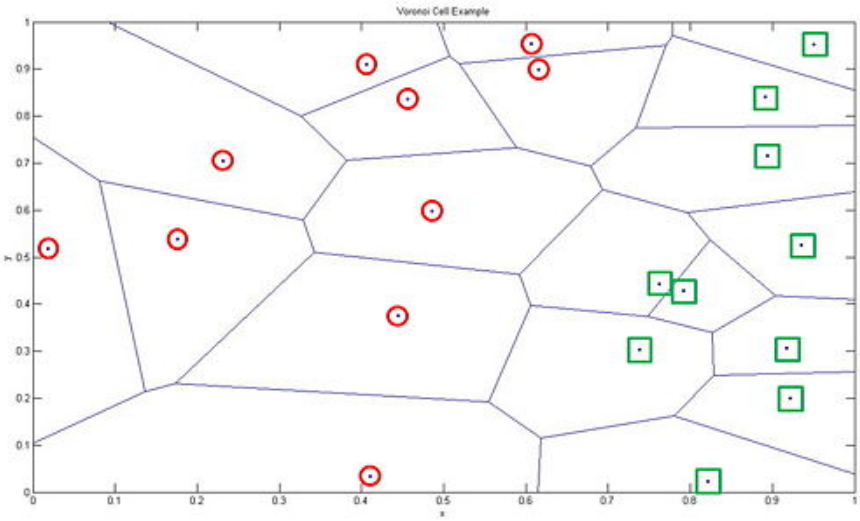
$$c(x_1 - a)^2 + d(x_2 - b)^2 - 1 = 0$$

$$\Rightarrow cx_1^2 + dx_2^2 - 2acx_1 - 2bdx_2 + (a^2c + b^2d - 1) = 0$$
$$\Rightarrow ((a^2c + b^2d - 1), -2ac, -2bd, 0, c, d).(1, x_1, x_2, x_1x_2, x_1^2, x_2^2) = 0$$

This shows that an elliptical boundary is linear in this feature space, allowing linear separability.

2. **Kernelization and its relation to k-nearest neighbor:** The nearest-neighbour classifier assigns a new input vector $\mathbf{x}$ to the same class as that of the nearest input vector $\mathbf{x}^n$ from the training set, where in the simplest case, the distance is defined by the Euclidean metric $||\mathbf{x} - \mathbf{x}^n||^2$.

   By expressing this rule in terms of scalar products and then making use of kernel substitution, formulate the nearest-neighbor classifier for a general nonlinear kernel.



(**Figure 2.** The Voronoi tessellation of 1-nearest neighbor on a sample dataset, from `https://www.projectrhea.org/rhea/index.php/Estimation_Using_Nearest_Neighbor`. The decision surface is a contiguous set of hyperplane segments between the +1 and -1 points.)

Let $D(\mathbf{x}, \mathbf{x}^n)$ represents the distance be between vector $\mathbf{x}$ and $\mathbf{x}^n$. The distance criterian for the nearest neighbor classifier can be expressed as follows:

$$D(\mathbf{x}, \mathbf{x}^n) = ||\mathbf{x} - \mathbf{x}^n||^2$$
$$= \mathbf{x}^T\mathbf{x} + [\mathbf{x}^n]^T\mathbf{x}^n - 2\mathbf{x}^T\mathbf{x}^n$$
$$= k(\mathbf{x}, \mathbf{x}) + k(\mathbf{x}^n\mathbf{x}^n) - 2k(\mathbf{x}, \mathbf{x}^n)$$

where, $k(\mathbf{x}^n\mathbf{x}^n) = [\mathbf{x}^n]^T\mathbf{x}^n$. Now, we can obtain non-linear kernel by replacing the linear kernel with some other choice of kernel function. Here, $\mathbf{x}^T$ represents the transpose of the vector $\mathbf{x}$.

3. **Kernelization:** Show that a diagonal matrix W whose elements satisfy $0 < W_{ii} < 1$ is positive definite. Futhermore, show that the sum of two positive semidefinite matrices is itself positive semidefinite.

   Suppose, W is a $n \times n$ matrix. Now, remember the definition of a positive definite matrix. For any n dimensional, non zero vector $\mathbf{x} \in \mathbb{R}^n$, we need to show that $\mathbf{x}^T W \mathbf{x} > 0$. Now, $\mathbf{x}^T W \mathbf{x} = \sum_{i=1}^n x_i^2 W_{ii}$.

Note that, $x_i \in \mathbb{R}$*, this* $x_i^2 \geq 0$ *where the equality holds only when* $x_i = 0$*. But according to the question* $\mathbf{x}$ *is a non-zero vector i.e atleast one element of the vector is not equal to* 0*. Thus* $\sum_{i=1}^{n} x_i^2 W_{ii} > 0$*. Hence* W *is a positive definite matrix.*

Next given two such positive definite matrix, W*,* W'*, we need to show that* $(W + W')$ *is also a positive definite matrix. Now according to the definition, for any* n *dimensional, non zero vector* $\mathbf{x}$*, we have* $\mathbf{x}^T W \mathbf{x} > 0$ *and* $\mathbf{x}^T W' \mathbf{x} > 0$*. Thus,*

$$\mathbf{x}^T (W + W')\mathbf{x} = \mathbf{x}^T W \mathbf{x} + \mathbf{x}^T W' \mathbf{x} > 0$$

Hence $(W + W')$ *is also a positive definite matrix.*